Raman Randive

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Professional Summary

Computer Engineering student specializing in **Artificial Intelligence, Machine Learning, and NLP** with experience in **RAG pipelines, multi-agent systems, A2A orchestration, and MCP-based tool integration**. Proficient in deep learning, embedding-based retrieval, and LLM-powered architectures using PyTorch and modern ML frameworks. Focused on building scalable, production-grade intelligent systems with strong foundations in model design and backend integration.

Education

Pimpri Chinchwad College of Engineering (PCCOE) — **Pune, India**
Bachelor of Technology in Computer Engineering ***(CGPA: 8.35)*** — *Aug. 2023 – June 2027*
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP, Machine Learning, Web Technologies

Technical Skills

AI/ML: Deep Learning, NLP, RAG Pipelines, Multi-Agent Systems, A2A Orchestration, MCP Tool, PyTorch, TensorFlow, LangChain, ChromaDB, Neural Networks
Programming: Python, TypeScript, JavaScript (ES6+), C/C++, SQL
Backend & Systems: Node.js, Express.js, REST APIs, WebSockets, Supabase, Firebase, neo4j
Web Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Databases: PostgreSQL, MongoDB, MySQL
Tools & Platforms: Git, GitHub, Linux (Ubuntu), Docker, Postman, Vercel, Render

Projects

Multi-Source AI Chatbot (PDF, CSV, Links, Web Scraping) | *Python, RAG, LangChain, ChromaDB, NLP* — Code

- Built a **Retrieval-Augmented Generation (RAG)** chatbot capable of querying structured and unstructured data from PDFs, CSV files, web links, and scraped content.
- Designed a modular ingestion pipeline using **CURL CFFI-based scraping**, document loaders, text chunking, and embedding-based vector indexing with ChromaDB.
- Implemented context-aware prompt orchestration and semantic retrieval to enable multi-document question answering.

Weapon Detection Model Training (Computer Vision) | *PyTorch, YOLOv8, OpenCV*

- Trained and fine-tuned a **YOLOv8-based object detection model** for weapon identification using curated datasets.
- Performed dataset preprocessing, augmentation, and hyperparameter tuning to improve detection robustness and generalization.
- Evaluated model performance using precision-recall metrics, mAP, and validation loss monitoring.

Research Experience

Comparative Study of Spatiotemporal Deep Learning Architectures for Video Deepfake Detection *(Under Review)*

- Conducted a systematic evaluation of **ResNet–LSTM, Temporal Attention, and GATv2-based Graph Neural Networks** for video-level deepfake detection on the FaceForensics++ dataset.
- Implemented 3D ResNet-18 spatiotemporal feature extraction with clip-level temporal modeling and achieved best performance using sequential modeling (AUC: 0.9344).

Experience

Technical Head – CESA ACM Student Chapter — **Aug 2025 – Present**

- Directed technical strategy and execution of AI/ML and full-stack initiatives, overseeing architecture, deployment, and scalability of student-led projects.
- Organized competitive coding contests and advanced workshops while mentoring 50+ students in DSA, system design, and applied machine learning.

Achievements

- **Competitive Programming:** Solved 400+ problems across Codeforces, LeetCode, and CodeChef in DSA.
- **CS50x – Harvard University (edX):** Completed rigorous training in algorithms, systems programming, and computational problem-solving.
- **ACM Winter School 2024:** Selected participant in advanced algorithmic design track.
- **Research & IP:** Presented research on spatiotemporal deepfake detection and secured copyright for **Safe Route Navigator**, a crime-aware geospatial routing system.